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Knowledge Check: What Are Neural Network Models?
Practice Assignment • 10 min

Due Oct 23, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the primary role of an activation function? 1 / 1 point

- (•) To introduce non-linearity into the model
- ( ) To count the number of layers in a network
- ( ) To normalize the input features
- ( ) To delete noisy input tokens

Nice work

Correct! Activation functions introduce non-linearity, allowing the model to learn complex patterns.

2. How does a leaky ReLU differ from a standard ReLU? 1 / 1 point

- ( ) It produces random outputs for negative inputs
- (•) It allows a small negative slope when the input is negative
- ( ) It scales all inputs equally
- ( ) It ignores all positive inputs

Nice work

Correct! Leaky ReLU has a small negative slope for negative inputs.

3. Which function is commonly used as an *activation function* in deep neural networks? 1 / 1 point

- ( ) Logarithmic function
- ( ) Fourier transform
- (•) Rectified Linear Unit (ReLU)
- ( ) Sorting function

Nice work

Correct! ReLU is a very common activation function in deep learning.

4. In the TLU formulation, what is the purpose of introducing a bias term? 1 / 1 point

- ( ) To ensure that weights remain constant
- ( ) To eliminate the need for activation functions
- ( ) To remove all input weights
- (•) To shift the activation threshold

Nice work

Correct! The bias shifts the activation threshold, helping the model learn patterns that don't pass through the origin.

5. What does a threshold logic unit (TLU) do? 1 / 1 point

- ( ) Computes the number of tokens in a sequence
- ( ) Sorts input tokens alphabetically
- ( ) Predicts future values in a time series
- (•) Combines weighted inputs and checks if the result exceeds a threshold

Nice work

Correct! A TLU combines weighted inputs and checks whether the result exceeds a threshold to produce an output.